

BEC303 Network Analysis — Module 2 Quiz

Network Theorems for DC and AC Circuits

Field	Value
Subject code	BEC303
Module	2 — Network Theorems for DC and AC Circuits
Duration	45 minutes
Maximum marks	50 (Part A 20 + Part B 30)
Attempts allowed	1
Passing marks	20
Textbook	W. H. Hayt, J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, 8th ed., 2012

Instructions

- Part A carries 20 one-mark multiple choice questions. Exactly one option is correct.
- Part B carries 6 descriptive questions totalling 30 marks.
- State explicitly which theorem you are invoking and why its hypotheses hold.
- Power must always be computed from total responses, never from partial ones.
- Non-programmable scientific calculators are permitted.

Course outcome and Bloom coverage

Band	Questions	Share	Target
L1–L2 Remember / Understand	8	24%	~30%
L3–L4 Apply / Analyse	20	61%	~45%
L5–L6 Evaluate / Create	5	15%	~25%

Course outcome spread: CO2: 33. Total questions: 33.

Part A — Multiple choice (20 × 1 mark = 20 marks)

A1. Superposition may be applied to compute which of the following? [CO2 | L2 | 1 mark | easy | Ch.5, pp.123-133]

- (A) Branch current
- (B) Power in a resistor
- (C) RMS value of a total waveform
- (D) Energy stored in a capacitor

A2. Two sources each drive 2 mA through a 1 k Ω resistor. The true power in the resistor is [CO2 | L3 | 1 mark | medium | Ch.5, pp.123-133]

- (A) 16 mW

- (B) 8 mW
- (C) 4 mW
- (D) 32 mW

A3. When applying superposition, an ideal independent current source is deactivated by [C02 | L1 | 1 mark | easy | Ch.5, pp.123-133]

- (A) replacing it with an open circuit
- (B) replacing it with a short circuit
- (C) setting its parallel resistance to zero
- (D) reversing its direction

A4. A practical 10 V source with $2\ \Omega$ internal resistance is deactivated for a superposition step. The branch becomes [C02 | L2 | 1 mark | medium | Ch.5, pp.123-133]

- (A) a plain $2\ \Omega$ resistance
- (B) a short circuit
- (C) an open circuit
- (D) a 10 V source with no resistance

A5. Sources at 0 rad/s, 377 rad/s and 1000 rad/s drive the same network. The responses must be combined [C02 | L4 | 1 mark | hard | Ch.5, pp.123-133]

- (A) in the time domain, after solving each frequency separately
- (B) as phasors, by vector addition
- (C) by taking the largest of the three
- (D) as RMS magnitudes in quadrature

A6. The reciprocity theorem requires the network to be [C02 | L2 | 1 mark | medium | TB2 §4.9]

- (A) linear, passive, bilateral, with zero initial conditions
- (B) linear only
- (C) linear and containing at least one dependent source
- (D) planar and resistive

A7. A 20 V source in branch A produces 4 mA in branch B of a passive ladder. Moving the source to branch B produces, in branch A, [C02 | L3 | 1 mark | medium | TB2 §4.9]

- (A) 4 mA
- (B) 20 mA
- (C) a value that cannot be found without re-solving
- (D) 2 mA

A8. Which network is non-reciprocal? [C02 | L3 | 1 mark | medium | TB2 §4.9]

- (A) A common-emitter amplifier small-signal model
- (B) An unbalanced resistive ladder
- (C) An LC low-pass filter
- (D) An ideal transformer

A9. The Thévenin voltage of a network is defined as [C02 | L1 | 1 mark | easy | Ch.5, pp.141-152]

- (A) the open-circuit voltage at the port
- (B) the voltage across the load when connected
- (C) the voltage across the internal resistance

- (D) the short-circuit current times the load resistance

A10. For a network containing dependent sources, R_{th} is best found by [C02 | L3 | 1 mark | medium | Ch.5, pp.141-152]

- (A) applying a test source at the dead port and taking V_{test}/I_{test}
- (B) deactivating all sources and reducing the resistive network
- (C) measuring the load resistance
- (D) summing all resistances in the network

A11. A 12 V source in series with 4 k Ω , with 6 k Ω to ground, has a Thévenin equivalent at the 6 k Ω of [C02 | L3 | 1 mark | medium | Ch.5, pp.141-152]

- (A) 7.2 V, 2.4 k Ω
- (B) 12 V, 10 k Ω
- (C) 7.2 V, 10 k Ω
- (D) 4.8 V, 2.4 k Ω

A12. The Norton current I_N is [C02 | L2 | 1 mark | easy | Ch.5, pp.141-152]

- (A) the short-circuit current at the port
- (B) V_{th} divided by the load resistance
- (C) the current through the load when connected
- (D) the sum of all source currents

A13. Maximum power is delivered to a DC load when [C02 | L2 | 1 mark | easy | Ch.5, pp.152-154]

- (A) $R_L = R_{th}$
- (B) $R_L \gg R_{th}$
- (C) $R_L \ll R_{th}$
- (D) $R_L = 2 \cdot R_{th}$

A14. At the maximum power transfer condition the efficiency of power delivery is [C02 | L3 | 1 mark | medium | Ch.5, pp.152-154]

- (A) 50%
- (B) 100%
- (C) 75%
- (D) undefined

A15. With $V_{th} = 12$ V and $R_{th} = 50$ Ω , the maximum power deliverable to a load is [C02 | L3 | 1 mark | medium | Ch.5, pp.152-154]

- (A) 0.72 W
- (B) 2.88 W
- (C) 1.44 W
- (D) 0.36 W

A16. For an AC source with $Z_{th} = 30 + j40$ Ω , maximum power is delivered when Z_L equals [C02 | L4 | 1 mark | hard | Ch.10, pp.397-406]

- (A) $30 - j40$ Ω
- (B) $30 + j40$ Ω
- (C) 50 Ω
- (D) $-30 + j40$ Ω

A17. In a $50\ \Omega$ RF system the load is matched primarily to [C02 | L4 | 1 mark | hard | Ch.5, pp.152-154]

- (A) prevent reflections on the transmission line
- (B) maximise efficiency
- (C) minimise the source current
- (D) simplify the Thévenin calculation

A18. An audio power amplifier is designed with output impedance far below the loudspeaker impedance in order to [C02 | L4 | 1 mark | hard | Ch.5, pp.152-154]

- (A) obtain high efficiency and a high damping factor
- (B) deliver maximum possible power to the speaker
- (C) satisfy the reciprocity conditions
- (D) make the Thévenin equivalent exact

A19. To confirm both parameters of a claimed Thévenin equivalent, you should check [C02 | L5 | 1 mark | medium | Ch.5, pp.141-152]

- (A) the open-circuit and the short-circuit cases
- (B) the open-circuit case only
- (C) the power balance only
- (D) the matched-load case only

A20. If a load is to be swept over five values, the cheapest route to the load power is [C02 | L5 | 1 mark | medium | Ch.5, pp.123-154]

- (A) compute one Thévenin equivalent, then five divider evaluations
- (B) five full nodal analyses
- (C) five superposition solves
- (D) five reciprocity swaps

Part B — Descriptive (13 questions, 30 marks)

B1. State the superposition theorem and explain, with an algebraic expansion, why it cannot be applied to power. [C02 | L4 | 3 marks | Ch.5, pp.123-133]

B2. A 10 V source in series with $2\ \text{k}\Omega$ and a 3 mA source both feed a common $6\ \text{k}\Omega$ load to ground. Find the load voltage by superposition and state each source's contribution. [C02 | L3 | 3 marks | Ch.5, pp.123-133]

B3. State the reciprocity theorem, list its four conditions, and give one example each of a reciprocal and a non-reciprocal network. [C02 | L3 | 3 marks | TB2 §4.9]

B4. A port consists of a $2\ \text{k}\Omega$ resistor in series with a CCVS of value $3000i_x$, where i_x is the port current. Find R_{th} and explain why the look-back method cannot be used. [C02 | L4 | 3 marks | Ch.5, pp.141-152]

B5. Derive the maximum power transfer condition for a DC source and state the efficiency at that condition. [C02 | L4 | 2 marks | Ch.5, pp.152-154]

B6. An antenna presents $Z_{th} = 50 + j30\ \Omega$. Choose the load for maximum power transfer and explain why magnitude matching is insufficient. [C02 | L5 | 2 marks | Ch.10, pp.397-406]

- B7.** Explain how an independent voltage source and an independent current source are deactivated, and what happens to an associated internal resistance. [C02 | L2 | 2 marks | Ch.5, pp.123-133]
- B8.** Sources at 0, 377 and 1000 rad/s drive the same network. Describe the correct solution procedure. [C02 | L4 | 2 marks | Ch.5, pp.123-133]
- B9.** A 12 V source in series with 4 k Ω has 6 k Ω to ground. Find the Thévenin equivalent at the 6 k Ω terminals and verify it with both limiting cases. [C02 | L3 | 2 marks | Ch.5, pp.141-152]
- B10.** Convert a Thévenin equivalent of 7.2 V with 2.4 k Ω to Norton form and verify with a 2.4 k Ω load. [C02 | L3 | 2 marks | Ch.5, pp.141-152]
- B11.** Two sources each drive 2 mA through a shared 1 k Ω resistor. Compute the power by both the naive and the correct routes and identify the discrepancy. [C02 | L4 | 2 marks | Ch.5, pp.123-133]
- B12.** Explain why maximum power transfer and maximum efficiency cannot be achieved simultaneously, and give one context in which each is the correct goal. [C02 | L5 | 2 marks | Ch.5, pp.152-154]
- B13.** A question asks for the power in a load as it is stepped through five values. Compare the cost of the Thévenin route with repeated direct analysis. [C02 | L5 | 2 marks | Ch.5, pp.123-154]

Answer key — Part A

Q	Answer	CO	Bloom	Reference
A1	A	CO2	L2	Ch.5, pp.123-133
A2	A	CO2	L3	Ch.5, pp.123-133
A3	A	CO2	L1	Ch.5, pp.123-133
A4	A	CO2	L2	Ch.5, pp.123-133
A5	A	CO2	L4	Ch.5, pp.123-133
A6	A	CO2	L2	TB2 §4.9
A7	A	CO2	L3	TB2 §4.9
A8	A	CO2	L3	TB2 §4.9
A9	A	CO2	L1	Ch.5, pp.141-152
A10	A	CO2	L3	Ch.5, pp.141-152
A11	A	CO2	L3	Ch.5, pp.141-152
A12	A	CO2	L2	Ch.5, pp.141-152
A13	A	CO2	L2	Ch.5, pp.152-154
A14	A	CO2	L3	Ch.5, pp.152-154
A15	A	CO2	L3	Ch.5, pp.152-154
A16	A	CO2	L4	Ch.10, pp.397-406
A17	A	CO2	L4	Ch.5, pp.152-154
A18	A	CO2	L4	Ch.5, pp.152-154

Q	Answer	CO	Bloom	Reference
A19	A	CO2	L5	Ch.5, pp.141-152
A20	A	CO2	L5	Ch.5, pp.123-154

Part A — why each option is right or wrong

A1. Correct answer: (A) Branch current

- (A) Correct — current is a linear response and therefore decomposes across sources.
- (B) Wrong - power is quadratic, so expanding the square leaves a cross term that superposition of powers discards.
- (C) Wrong - RMS involves squaring, so it inherits the same quadratic problem as power.
- (D) Wrong - stored energy is proportional to v squared and is therefore not a linear response.

A2. Correct answer: (A) 16 mW

- (A) Correct — total current is 4 mA, so $p = (0.004)^2 \cdot 1000 = 16$ mW.
- (B) Wrong - 8 mW is the naive sum of partial powers, which discards the 8 mW cross term.
- (C) Wrong - 4 mW is one partial power alone.
- (D) Wrong - this doubles the correct answer, as if the currents were 8 mA.

A3. Correct answer: (A) replacing it with an open circuit

- (A) Correct — zero current through a branch is exactly an open circuit.
- (B) Wrong - shorting deactivates a voltage source; shorting a current source forces its full current through the short.
- (C) Wrong - any associated resistance stays in the circuit unchanged.
- (D) Wrong - reversing a source changes the excitation rather than removing it.

A4. Correct answer: (A) a plain $2\ \Omega$ resistance

- (A) Correct — only the ideal source part is zeroed; the internal resistance belongs to the network.
- (B) Wrong - this deletes the internal resistance and changes every parallel combination downstream.
- (C) Wrong - an open circuit would remove the branch entirely, which is the deactivation rule for a current source.
- (D) Wrong - this leaves the excitation live, which is the opposite of deactivation.

A5. Correct answer: (A) in the time domain, after solving each frequency separately

- (A) Correct — phasors at different frequencies have no fixed relative phase, so only the time-domain waveforms may be added.
- (B) Wrong - vector addition of phasors is valid only within a single frequency.
- (C) Wrong - this simply discards two of the three contributions.
- (D) Wrong - quadrature addition applies to RMS power contributions, not to the instantaneous response.

A6. Correct answer: (A) linear, passive, bilateral, with zero initial conditions

- (A) Correct — all four conditions are needed together; dropping any one voids the guarantee.
- (B) Wrong - linearity alone suffices for superposition and Thevenin but not for reciprocity.
- (C) Wrong - a dependent source is precisely what destroys bilaterality and breaks the theorem.

- (D) Wrong - planarity is a requirement of mesh analysis, not of reciprocity, and reactive elements are permitted.

A7. Correct answer: (A) 4 mA

- (A) Correct — the transfer resistance of 5 k Ω is symmetric, so the same 4 mA appears.
- (B) Wrong - this divides the source value by 1 kilohm for no stated reason.
- (C) Wrong - reciprocity exists exactly so that the second configuration need not be re-solved.
- (D) Wrong - halving the current has no basis; reciprocity preserves the value exactly.

A8. Correct answer: (A) A common-emitter amplifier small-signal model

- (A) Correct — the current-controlled current source transmits forwards only, so $Z_{12} \neq Z_{21}$.
- (B) Wrong - physical asymmetry does not break reciprocity; a passive ladder is always reciprocal.
- (C) Wrong - an LC filter is passive and bilateral and is therefore reciprocal.
- (D) Wrong - an ideal transformer transmits equally in both directions and is reciprocal.

A9. Correct answer: (A) the open-circuit voltage at the port

- (A) Correct — the load must be removed, or the divider is loaded and the value comes out low.
- (B) Wrong - this is the loaded terminal voltage, always less than V_{th} for a positive R_{th} .
- (C) Wrong - the internal resistance drop is what separates V_{th} from the loaded voltage.
- (D) Wrong - I_{sc} times R_{th} gives V_{th} ; using the load resistance instead is a different and wrong quantity.

A10. Correct answer: (A) applying a test source at the dead port and taking V_{test}/I_{test}

- (A) Correct — dependent sources may not be deactivated, so the test-source method is the only generally safe route.
- (B) Wrong - deactivating a dependent source changes the network and gives a wrong R_{th} , often by tens of percent.
- (C) Wrong - R_{th} is a property of the source network and has nothing to do with the load.
- (D) Wrong - resistances only add in series; a network sum has no meaning.

A11. Correct answer: (A) 7.2 V, 2.4 k Ω

- (A) Correct — $V_{th} = 12 \cdot 6/10 = 7.2$ V and $R_{th} = 4k \parallel 6k = 2.4$ k Ω .
- (B) Wrong - this ignores the divider and adds the resistors in series instead of in parallel.
- (C) Wrong - the voltage is right but R_{th} uses a series rather than a parallel combination.
- (D) Wrong - 4.8 V comes from using the 4 kilohm rather than the 6 kilohm in the divider numerator.

A12. Correct answer: (A) the short-circuit current at the port

- (A) Correct — I_N is a property of the network alone, measured with the terminals shorted.
- (B) Wrong - dividing by the load brings the load into a quantity that must be load-independent.
- (C) Wrong - the load current depends on the load and is not the Norton parameter.
- (D) Wrong - source currents redistribute through the network and do not simply add at the port.

A13. Correct answer: (A) $R_L = R_{th}$

- (A) Correct — differentiating $p = V_{th}^2 R_L / (R_{th} + R_L)^2$ and setting the result to zero gives $R_L = R_{th}$.
- (B) Wrong - this maximises efficiency, not delivered power, and the delivered power tends to zero.
- (C) Wrong - this maximises current but the load voltage collapses, so delivered power again tends to zero.
- (D) Wrong - a 2:1 mismatch delivers about 89 percent of the maximum, not the maximum itself.

A14. Correct answer: (A) 50%

- (A) Correct — R_L equals R_{th} , so the two dissipate equally and half the total power is lost internally.
- (B) Wrong - 100 percent efficiency would require zero internal resistance, which contradicts the matched condition.
- (C) Wrong - 75 percent corresponds to $R_L = 3 R_{th}$, a deliberate mismatch.
- (D) Wrong - efficiency is perfectly well defined as the ratio of load power to total power.

A15. Correct answer: (A) 0.72 W

- (A) Correct — $p_{max} = V_{th}^2 / (4R_{th}) = 144 / 200 = 0.72$ W.
- (B) Wrong - this omits the factor of 4 and computes V_{th} squared over R_{th} .
- (C) Wrong - this uses a factor of 2 instead of 4 in the denominator.
- (D) Wrong - this halves the correct answer, as if only a quarter of the power reached the load.

A16. Correct answer: (A) $30 - j40 \Omega$

- (A) Correct — the conjugate match cancels the source reactance and leaves a purely resistive loop.
- (B) Wrong - this doubles the loop reactance instead of cancelling it, applying the DC rule to an AC problem.
- (C) Wrong - matching the magnitude leaves the reactances uncanceled and falls short of the maximum.
- (D) Wrong - a negative resistance is not realisable in a passive load and does not follow from the conjugate rule.

A17. Correct answer: (A) prevent reflections on the transmission line

- (A) Correct — a mismatch returns energy down the line and corrupts the signal, which matters more than raw power transfer.
- (B) Wrong - a matched termination is only 50 percent efficient, so efficiency cannot be the motive.
- (C) Wrong - source current is not a design target in a transmission-line interface.
- (D) Wrong - calculation convenience never drives a physical impedance standard.

A18. Correct answer: (A) obtain high efficiency and a high damping factor

- (A) Correct — a deliberate mismatch trades peak deliverable power for efficiency and cone control.
- (B) Wrong - maximum power would require matching the speaker impedance, which is exactly what is avoided.
- (C) Wrong - reciprocity is irrelevant here and an amplifier is non-reciprocal by design.
- (D) Wrong - the Thevenin equivalent is exact for any linear network regardless of the impedance ratio.

A19. Correct answer: (A) the open-circuit and the short-circuit cases

- (A) Correct — two parameters require two independent conditions, and these are the cheapest pair.
- (B) Wrong - the open-circuit case confirms V_{th} and leaves R_{th} entirely untested.
- (C) Wrong - a power balance can be satisfied by a wrong pair of parameters that happen to be self-consistent.
- (D) Wrong - a single load value gives one equation for two unknowns.

A20. Correct answer: (A) compute one Thévenin equivalent, then five divider evaluations

- (A) Correct — the equivalent costs two solves once, after which each load value is trivial.
- (B) Wrong - this repeats the full network solution five times and ignores that only one branch changes.
- (C) Wrong - superposition is more expensive still, requiring one solve per source per load value.
- (D) Wrong - reciprocity swaps excitation and response; it does not help with a varying load.

Part B — model answers and marking scheme

B1 (3 marks) — CO2, L4, Ch.5, pp.123-133

Question. State the superposition theorem and explain, with an algebraic expansion, why it cannot be applied to power.

Model answer. Superposition states that in a linear circuit the voltage or current at any point equals the algebraic sum of the responses to each independent source acting alone, with other independent sources deactivated and all dependent sources left active. It fails for power because power is quadratic: $(i_1 + i_2)^2 R = i_1^2 R + i_2^2 R + 2i_1 i_2 R$. Superposing powers keeps the first two terms and discards the cross term. If $i_1 = i_2$ the true power is $4i_1^2 R$ against a naive $2i_1^2 R$ — a 100 % error.

Point required	Marks
Correct statement including the treatment of dependent sources	1
Algebraic expansion identifying the cross term	1
Quantified size of the error	1
Total	3

B2 (3 marks) — CO2, L3, Ch.5, pp.123-133

Question. A 10 V source in series with 2 k Ω and a 3 mA source both feed a common 6 k Ω load to ground. Find the load voltage by superposition and state each source's contribution.

Model answer. With the 3 mA source opened, $v' = 10 \times 6/(2+6) = 7.5$ V. With the 10 V source shorted, the 2 k Ω and 6 k Ω are in parallel giving 1.5 k Ω , so $v'' = 3 \text{ mA} \times 1.5 \text{ k}\Omega = 4.5$ V. Both have the same polarity, so $v = 12$ V. The voltage source contributes 62.5 % and the current source 37.5 % — attribution a single nodal solve would not reveal.

Point required	Marks
Both partial responses correct with the deactivations justified	2
Correct algebraic sum and attribution	1
Total	3

B3 (3 marks) — CO2, L3, TB2 §4.9

Question. State the reciprocity theorem, list its four conditions, and give one example each of a reciprocal and a non-reciprocal network.

Model answer. Reciprocity states that in a linear, passive, bilateral network with a single excitation, interchanging excitation and response leaves their ratio unchanged. The four conditions are linearity, passivity with only that single excitation, bilateral elements, and zero initial conditions. An unbalanced resistive ladder is reciprocal. A common-emitter small-signal model is non-reciprocal, because its dependent source transmits forward only, so $z_{12} \neq z_{21}$.

Point required	Marks
Correct statement of the theorem	1
All four conditions listed	1

Point required	Marks
One valid example of each type with a reason	1
Total	3

B4 (3 marks) — CO2, L4, Ch.5, pp.141-152

Question. A port consists of a 2 k Ω resistor in series with a CCVS of value 3000·i_x, where i_x is the port current. Find R_{th} and explain why the look-back method cannot be used.

Model answer. The look-back method requires deactivating all sources, but a dependent source may not be deactivated: its value is set by the circuit, so removing it changes the network. Use a test source instead. With V_{test} = 1 V, 1 = 2000·i_x + 3000·i_x = 5000·i_x, so i_x = 0.2 mA and R_{th} = 5 k Ω . Naive deactivation gives 2 k Ω , a 60 % error.

Point required	Marks
Correct reason why deactivation is impermissible	1
Correct test-source equation and R _{th}	2
Total	3

B5 (2 marks) — CO2, L4, Ch.5, pp.152-154

Question. Derive the maximum power transfer condition for a DC source and state the efficiency at that condition.

Model answer. For V_{th} with resistance R_{th} feeding R_L, $p = V_{th}^2 \cdot R_L / (R_{th} + R_L)^2$. Setting $dp/dR_L = 0$ gives $(R_{th} + R_L)^2 = 2R_L(R_{th} + R_L)$, hence $R_L = R_{th}$ and $p_{max} = V_{th}^2 / (4R_{th})$. Because the two resistances are equal they dissipate equally, so efficiency is exactly 50 %.

Point required	Marks
Correct power expression and differentiation	1
Correct condition, p _{max} and the 50 % efficiency statement	1
Total	2

B6 (2 marks) — CO2, L5, Ch.10, pp.397-406

Question. An antenna presents Z_{th} = 50 + j30 Ω . Choose the load for maximum power transfer and explain why magnitude matching is insufficient.

Model answer. The load must be the complex conjugate, Z_L = 50 – j30 Ω . The –j30 cancels the source reactance, leaving a purely resistive 100 Ω loop, so the delivered power reaches the available maximum $|V_{th}|^2 / (4 \times 50)$. Setting |Z_L| = |Z_{th}| instead leaves the reactances adding rather than cancelling, so loop current falls and delivered power falls short of the maximum.

Point required	Marks
Correct conjugate load	1
Correct explanation of why magnitude matching fails	1
Total	2

B7 (2 marks) — CO2, L2, Ch.5, pp.123-133

Question. Explain how an independent voltage source and an independent current source are deactivated, and what happens to an associated internal resistance.

Model answer. A voltage source is deactivated by setting its voltage to zero, which is a short circuit. A current source is deactivated by setting its current to zero, which is an open circuit. Any resistance associated with a practical source is part of the network, not the source, and remains in place — a 10 V source with $2\ \Omega$ internal resistance deactivates to a plain $2\ \Omega$, not to a wire.

Point required	Marks
Both deactivation rules correct	1
Correct treatment of the internal resistance	1
Total	2

B8 (2 marks) — CO2, L4, Ch.5, pp.123-133

Question. Sources at 0, 377 and 1000 rad/s drive the same network. Describe the correct solution procedure.

Model answer. Because impedance depends on ω , no single phasor domain can hold all three sources, so superposition is mandatory. Solve each frequency separately with impedances evaluated at that ω , treating the DC source as $\omega = 0$ with inductors shorted and capacitors opened. Convert each partial answer back to the time domain, then add the three time functions. Adding the phasors directly is meaningless because phasors at different frequencies have no fixed relative phase.

Point required	Marks
Correct per-frequency procedure including the DC case	1
Correct statement that recombination must be in the time domain, with the reason	1
Total	2

B9 (2 marks) — CO2, L3, Ch.5, pp.141-152

Question. A 12 V source in series with $4\ \text{k}\Omega$ has $6\ \text{k}\Omega$ to ground. Find the Thévenin equivalent at the $6\ \text{k}\Omega$ terminals and verify it with both limiting cases.

Model answer. V_{th} is the open-circuit voltage: $12 \times 6/(4+6) = 7.2\ \text{V}$. Deactivating the source, $R_{\text{th}} = 4\ \text{k}\Omega \parallel 6\ \text{k}\Omega = 2.4\ \text{k}\Omega$. Open-circuit check: the equivalent gives 7.2 V, matching the divider. Short-circuit check: the equivalent gives $7.2/2400 = 3\ \text{mA}$, and the original with terminals shorted gives $12/4000 = 3\ \text{mA}$. Both limits agree, confirming both parameters.

Point required	Marks
Correct V_{th} and R_{th}	1
Both limiting-case checks performed correctly	1
Total	2

B10 (2 marks) — CO2, L3, Ch.5, pp.141-152

Question. Convert a Thévenin equivalent of 7.2 V with $2.4\ \text{k}\Omega$ to Norton form and verify with a $2.4\ \text{k}\Omega$ load.

Model answer. $I_N = V_{th}/R_{th} = 7.2/2400 = 3 \text{ mA}$ and $R_N = R_{th} = 2.4 \text{ k}\Omega$, so the Norton form is 3 mA in parallel with 2.4 k Ω . With a 2.4 k Ω load the current divides equally, so the load takes 1.5 mA and develops $1.5 \text{ mA} \times 2.4 \text{ k}\Omega = 3.6 \text{ V}$. The Thévenin form gives $7.2 \times 2.4/(2.4+2.4) = 3.6 \text{ V}$, as it must.

Point required	Marks
Correct I_N and R_N	1
Correct verification against the Thévenin form	1
Total	2

B11 (2 marks) — CO2, L4, Ch.5, pp.123-133

Question. Two sources each drive 2 mA through a shared 1 k Ω resistor. Compute the power by both the naive and the correct routes and identify the discrepancy.

Model answer. Naive: $p_1 = p_2 = (0.002)^2 \times 1000 = 4 \text{ mW}$, giving 8 mW. Correct: total current is 4 mA, so $p = (0.004)^2 \times 1000 = 16 \text{ mW}$. The discrepancy is the cross term $2 \cdot i_1 \cdot i_2 \cdot R = 2 \times 0.002 \times 0.002 \times 1000 = 8 \text{ mW}$, which happens here to equal the entire naive answer.

Point required	Marks
Both values computed correctly	1
Cross term identified and quantified	1
Total	2

B12 (2 marks) — CO2, L5, Ch.5, pp.152-154

Question. Explain why maximum power transfer and maximum efficiency cannot be achieved simultaneously, and give one context in which each is the correct goal.

Model answer. At the matched condition $R_L = R_{th}$ the source and load dissipate equally, capping efficiency at 50 %. Efficiency rises toward 100 % only as $R_L \gg R_{th}$, where delivered power falls toward zero. Maximum power is the right goal when signal power is the scarce resource, as at an antenna or transducer input. Maximum efficiency is the right goal in power delivery, which is why a mains supply and an audio output stage both run with source impedance far below the load.

Point required	Marks
Correct argument that the two goals conflict	1
One valid context for each goal	1
Total	2

B13 (2 marks) — CO2, L5, Ch.5, pp.123-154

Question. A question asks for the power in a load as it is stepped through five values. Compare the cost of the Thévenin route with repeated direct analysis.

Model answer. The Thévenin route costs two network solves — one for V_{th} and one for R_{th} — after which each load value costs a single divider evaluation. Direct analysis costs one full network solve per load value, so five solves here. The equivalent breaks even at about three load values and wins by a widening margin beyond that. The saving comes from recognising that only one branch changes between cases.

Point required	Marks
Correct cost accounting for both routes	1
Correct break-even conclusion with the reason	1
Total	2